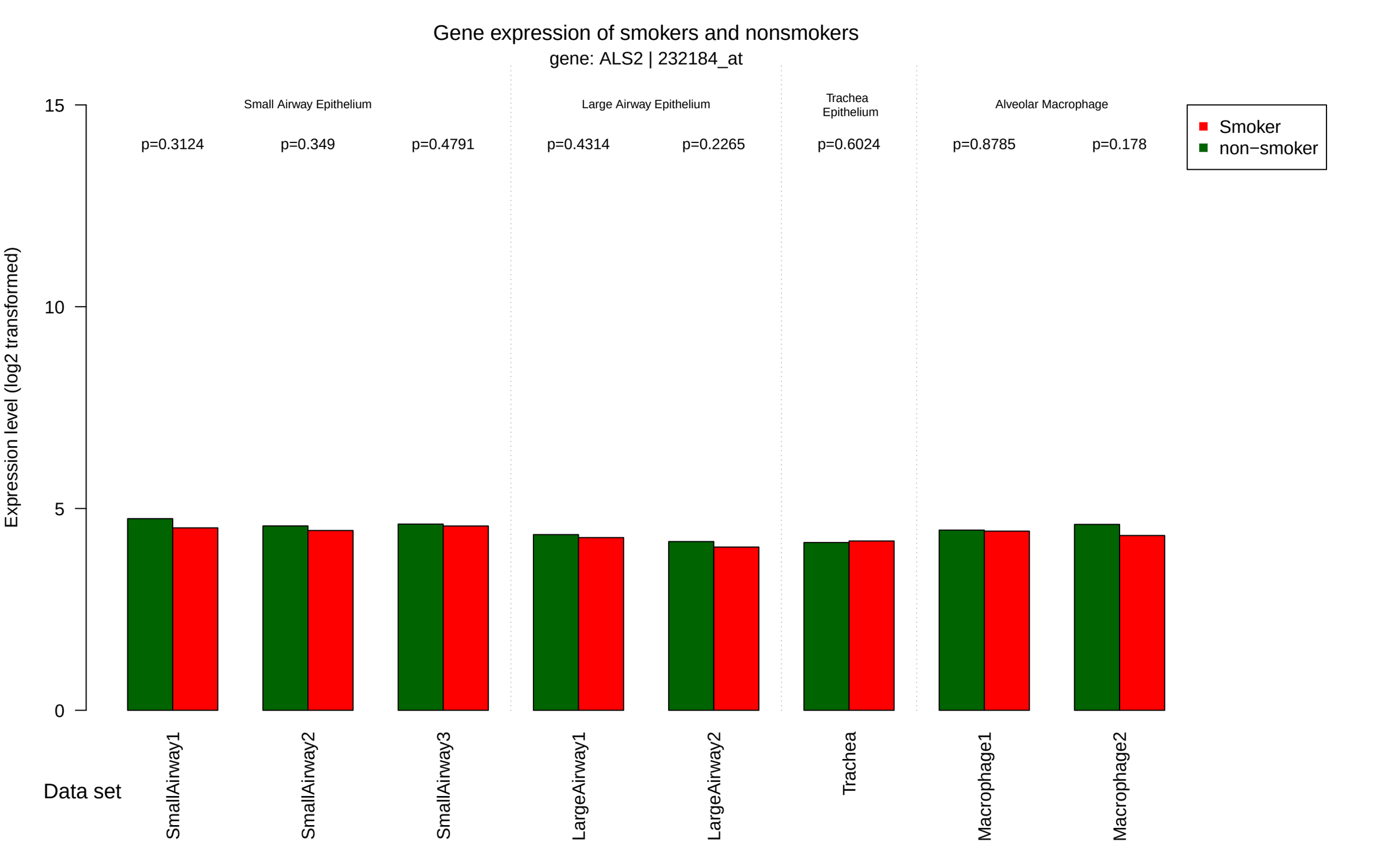
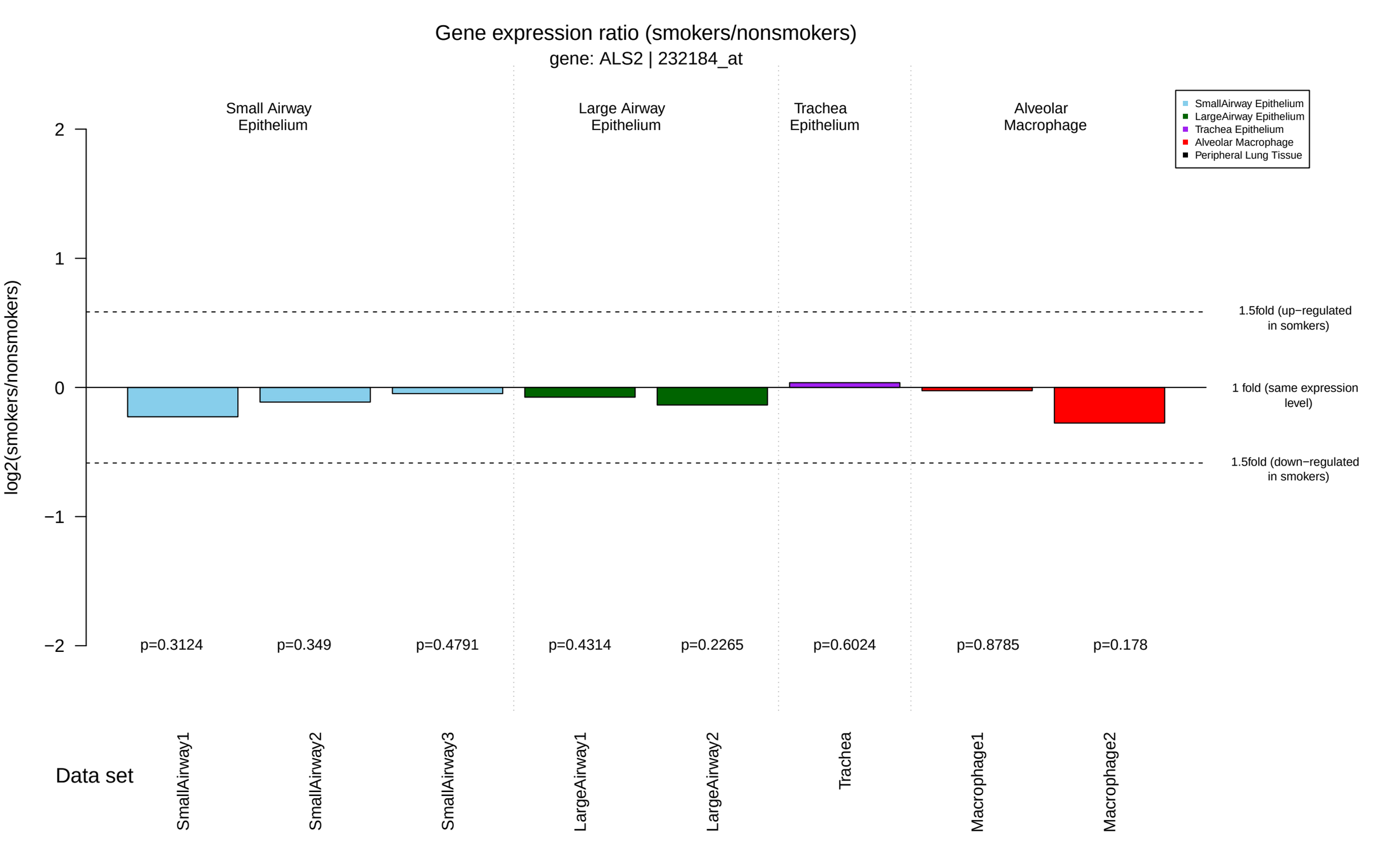
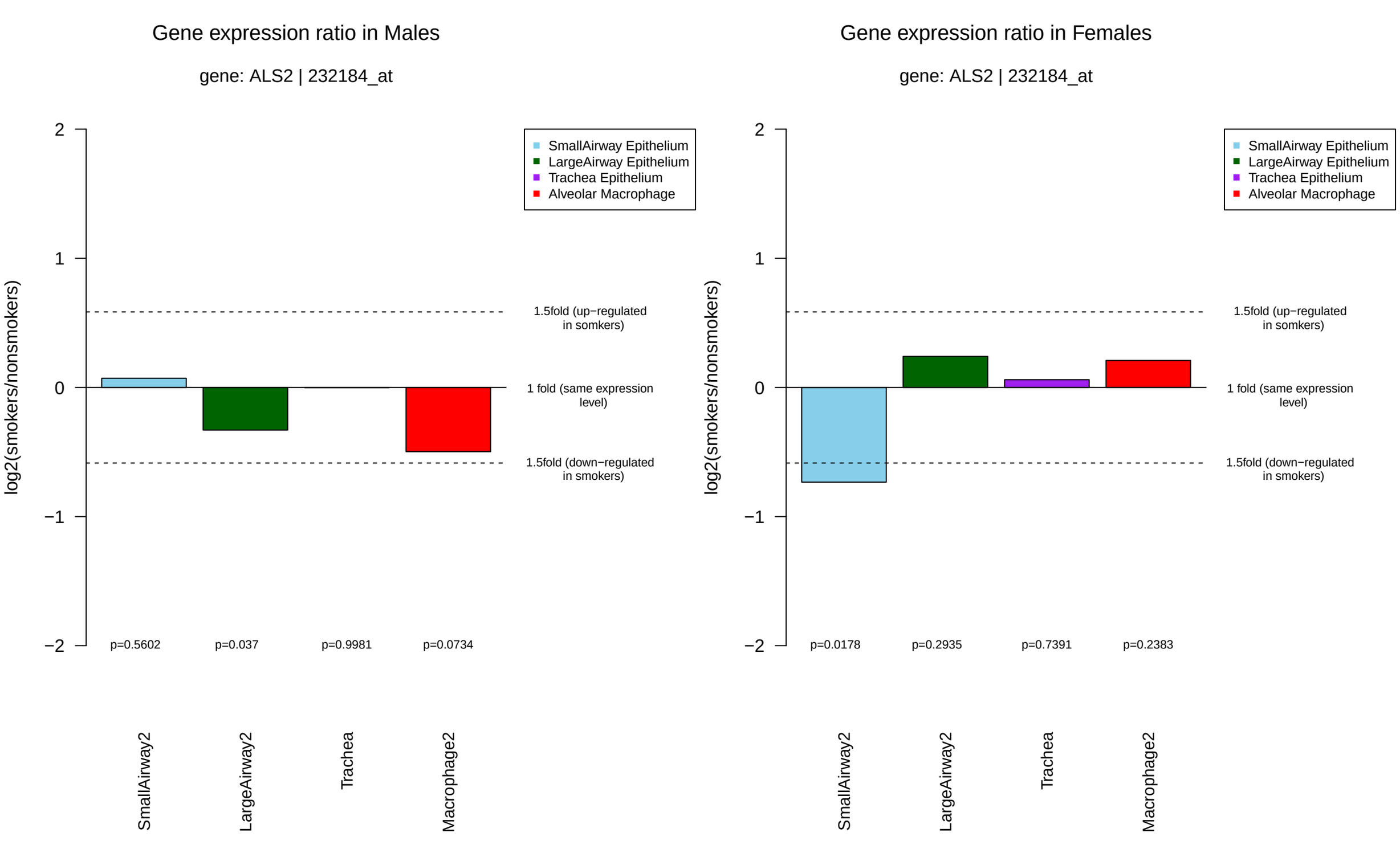
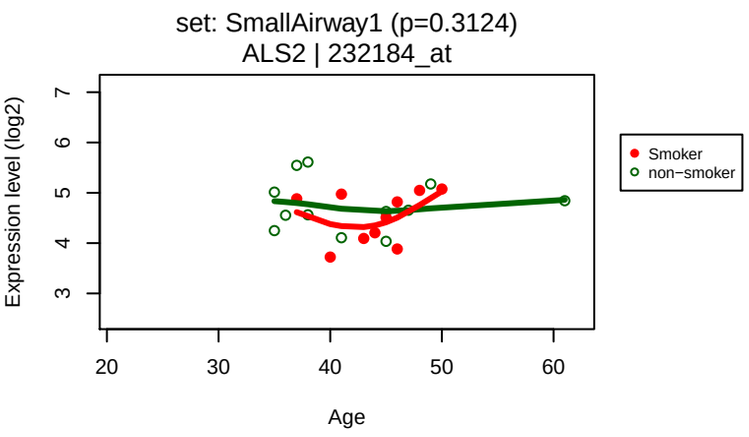
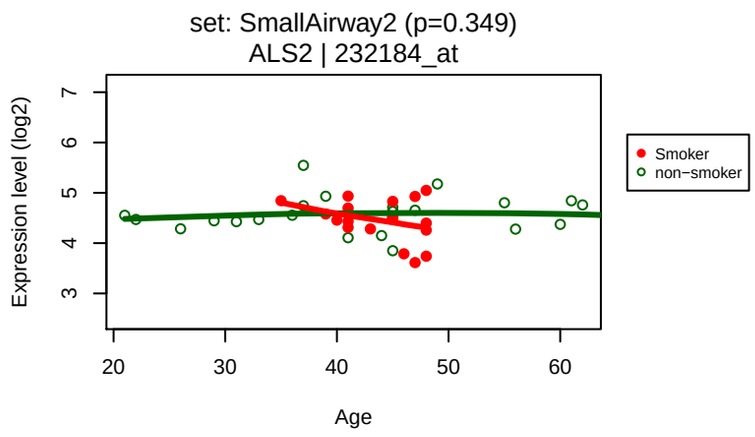
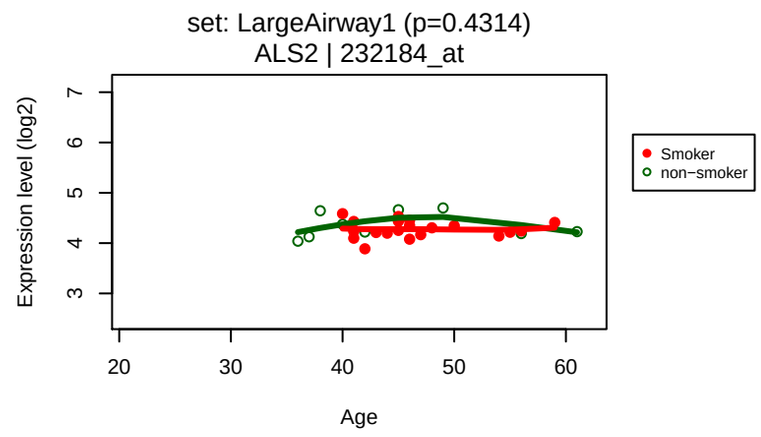

Gene expression of smokers and nonsmokers
gene: ALS2 | 232184_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3124
p=0.349
p=0.4791
p=0.4314
p=0.2265
p=0.6024
p=0.8785
p=0.178
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

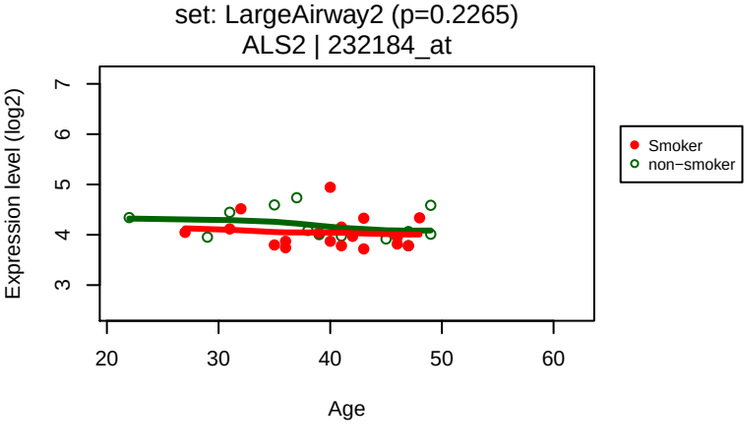

Gene expression ratio (smokers/nonsmokers)
gene: ALS2 | 232184_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3124
p=0.349
p=0.4791
p=0.4314
p=0.2265
p=0.6024
p=0.8785
p=0.178
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

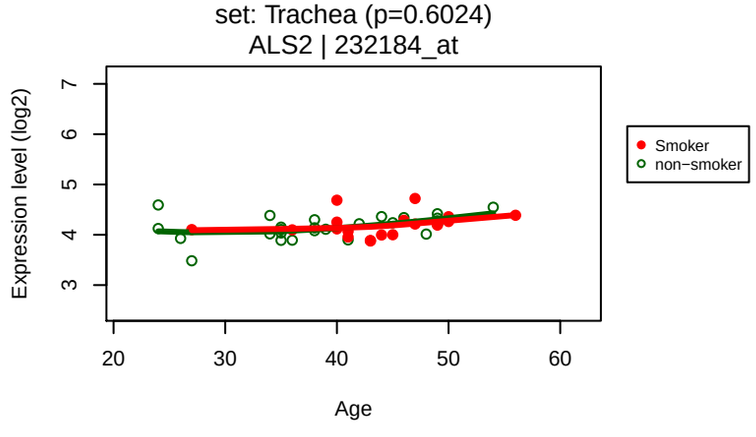

Gene expression ratio in Males
gene: ALS2 | 232184_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5602
p=0.037
p=0.9981
p=0.0734
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALS2 | 232184_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0178
p=0.2935
p=0.7391
p=0.2383
SmallAirway2
LargeAirway2
Trachea
Macrophage2

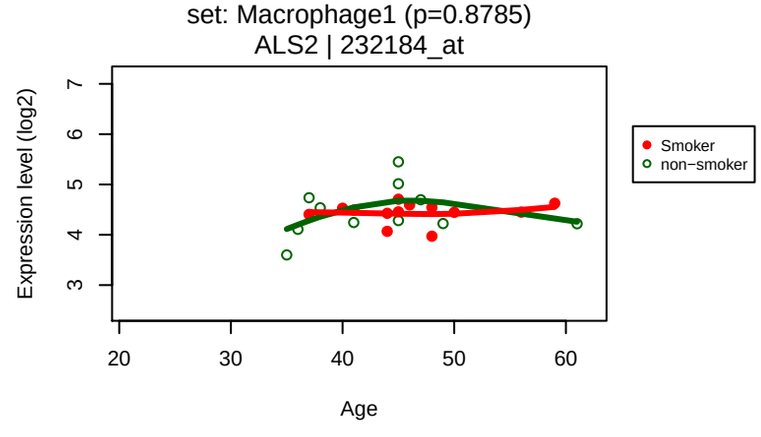

set: SmallAirway1 (p=0.3124)
ALS2 | 232184_at
Expression level (log2)
Age
Smoker
non−smoker

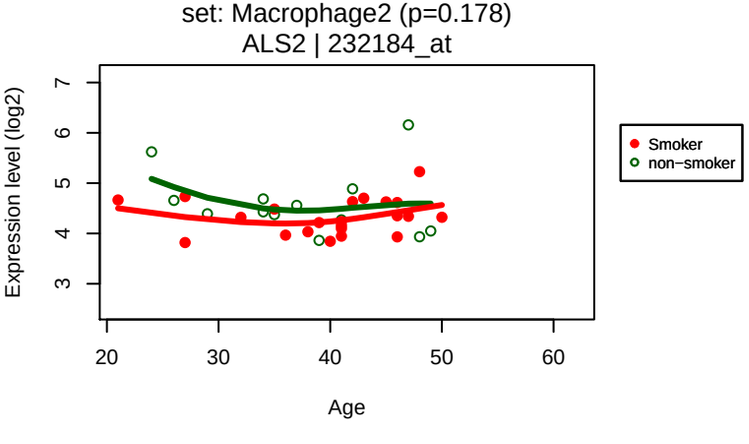

set: SmallAirway2 (p=0.349)
ALS2 | 232184_at
Expression level (log2)
Age
Smoker
non−smoker

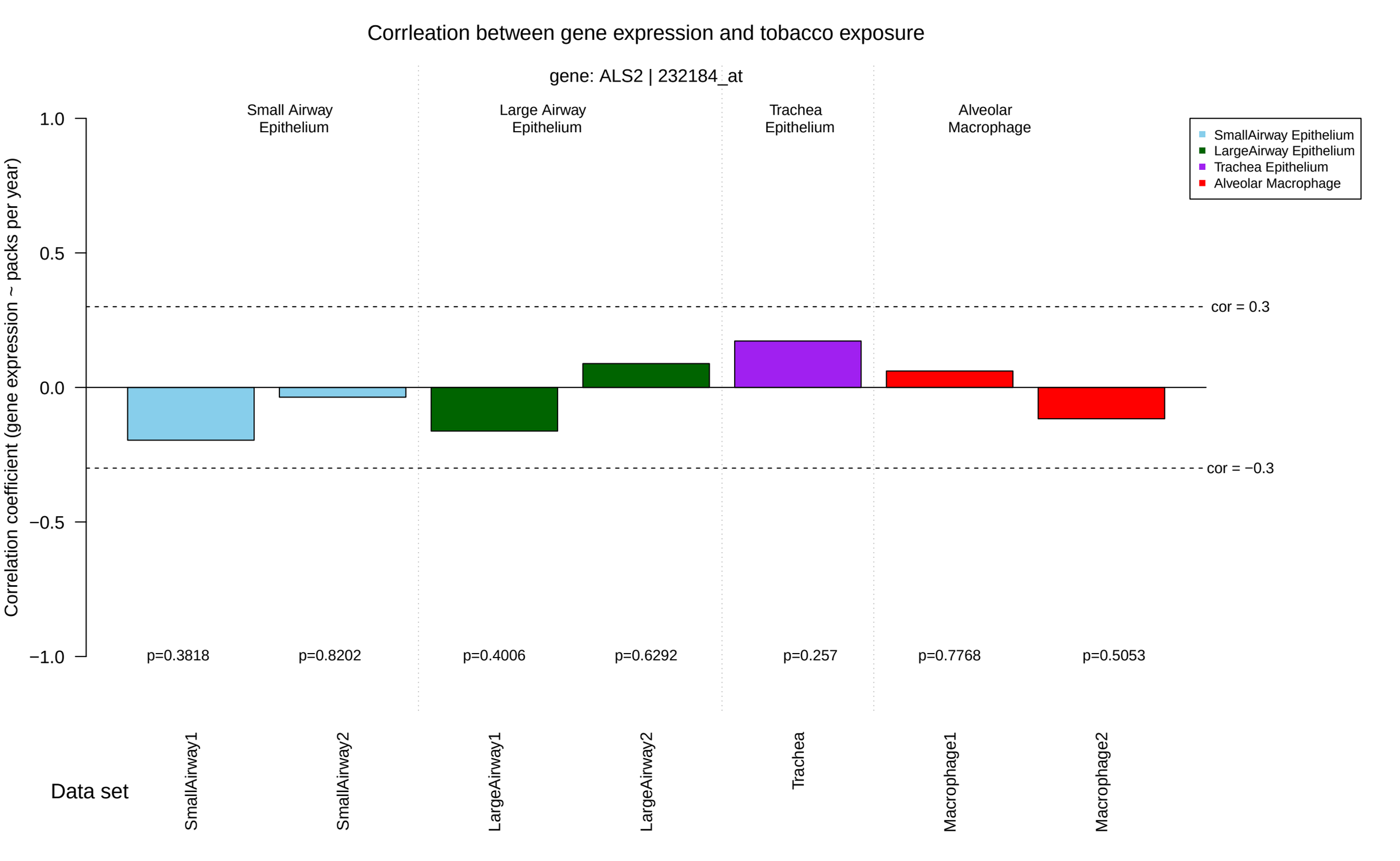

set: LargeAirway1 (p=0.4314)
ALS2 | 232184_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2265)
ALS2 | 232184_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6024)
ALS2 | 232184_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8785)
ALS2 | 232184_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.178)
ALS2 | 232184_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALS2 | 232184_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3818
p=0.8202
p=0.4006
p=0.6292
p=0.257
p=0.7768
p=0.5053
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2